

MICHAEL MAHARJAN

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SUMMARY

Healthcare-focused Software Engineer with 10 years of comprehensive experience in designing and implementing scalable, secure web applications and backend services. Proficient in **Python**, **Django**, **FastAPI**, and modern **JavaScript** frameworks such as **React.js**, with a proven track record in building RESTful APIs, integrating advanced authentication mechanisms, and optimizing database interactions for enhanced performance and reliability. Adept in deploying cloud-based architectures, implementing CI/CD pipelines, and following Agile methodologies to deliver innovative, user-centric solutions. Passionate about leveraging technology to improve healthcare outcomes through secure, efficient, and scalable systems.

ADDITIONAL INFORMATION

- **Languages & Frameworks:** Python, Django, FastAPI, Flask, React.js, Node.js, Express.js, C#, ASP.NET, .NET Core
 - **Database Management:** PostgreSQL, MySQL, MongoDB, Redis
 - **Cloud & DevOps:** AWS (EC2, S3, RDS, Lambda), Azure (DevOps, Synapse), Kubernetes, Jenkins, Docker
 - **Testing & Automation:** Pytest, Postman, Selenium, Puppeteer
 - **Tools & Methodologies:** Agile, CI/CD, Alembic, OAuth2, JWT, Tableau
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WORK EXPERIENCE

Senior Software Engineer, Oscar Health

January 2022 - Present

- Spearheaded the development of **Python**-based microservices using **FastAPI**, enabling seamless real-time patient data exchange while ensuring robust security through the integration of **OAuth2** and **JWT**-based authentication.
- Designed and deployed a serverless pipeline for processing OCR of patient referrals with an accuracy of ~98%, significantly reducing manual processing time by 99.95% and cutting patient onboarding time from 72 hours to just 2 minutes.
- Enhanced system performance and scalability by integrating **Redis** for caching and session management, achieving a 40% improvement in data processing speeds and overall responsiveness.
- Configured and maintained **Jenkins** pipelines for continuous integration and deployment, successfully automating complex workflows and reducing deployment times across all environments.
- Implemented comprehensive testing strategies using **Pytest** and **Postman**, covering unit, integration, and API validation tests to ensure high-quality code and reliable application endpoints.
- Collaborated with cross-functional teams to deliver a provider credentialing platform, reducing credentialing times by 90% while ensuring compliance with healthcare regulations.
- Built custom analytics tools to visualize patient data trends, enabling the management team to make data-driven decisions that improved operational efficiency by 30%.
- Developed a cloud-native **FastAPI** application deployed on **AWS**, utilizing S3 for secure document storage and Lambda for automated processing of incoming healthcare data.

Software Engineer, UnitedHealth Group

March 2019 - December 2021

- Built RESTful APIs with **Django** and seamlessly integrated them with **React.js** front-end applications, utilizing **Redux** for state management to deliver highly interactive and dynamic user experiences.
- Enhanced application performance by optimizing database interactions with **Django ORM** and **PostgreSQL**, achieving a 30% reduction in query execution times and improving scalability.
- Implemented robust authentication and security measures using token-based systems such as **JWT** and **OAuth2**, ensuring compliance with healthcare data security standards.
- Partnered with data scientists and healthcare professionals to build custom dashboards using **Tableau**, enabling stakeholders to visualize and analyze patient outcomes effectively.
- Designed CI/CD pipelines in **Azure DevOps** to automate deployment processes, reducing manual intervention and increasing deployment frequency with minimal errors.
- Automated claims processing workflows using serverless architectures, reducing manual intervention by 75% and improving data accuracy for billing processes.
- Developed high-performance backend systems to handle real-time analytics for large-scale healthcare datasets, ensuring uptime and reliability for mission-critical applications.

Full Stack Developer, Flatiron Health

September 2016 - March 2019

- Developed and maintained full-stack web applications using the **MEAN** stack, delivering robust and scalable solutions for oncology data management systems.
- Designed and implemented RESTful APIs using **Node.js** and **Express.js**, ensuring efficient data exchange between front-end and back-end services while leveraging **MongoDB** and **SQL** databases for data storage.
- Created intuitive and responsive user interfaces using **AngularJS** and Bootstrap, enhancing usability and accessibility for healthcare professionals across devices.
- Analyzed and processed large healthcare datasets using **Python** and **SQL**, creating actionable insights that improved decision-making and operational efficiency.
- Built automated reporting tools using **Tableau** and **Python** libraries such as **Matplotlib** and **Pandas**, presenting critical metrics to stakeholders in visually engaging formats.
- Implemented a real-time notification system for patient data updates, reducing delays in communication between healthcare providers by 50%.

Junior Software Developer, Cerner Corporation

April 2014 - August 2016

- Assisted in the development and enhancement of **.NET**-based web applications tailored for healthcare scheduling and billing systems, ensuring compliance with industry regulations.
- Implemented CRUD functionality using Entity Framework, streamlining database interactions with **SQL** Server and improving system reliability.
- Collaborated with senior developers to migrate legacy systems into modern web-based platforms, reducing operational costs by 20% and improving maintainability.
- Participated in **Agile** sprints, contributing to requirements gathering, technical documentation, and system design to align with client expectations.
- Conducted rigorous testing and debugging of applications to identify and resolve critical issues, ensuring smooth deployments and high-quality releases.
- Enhanced application security by implementing role-based access control (RBAC) and data encryption, improving compliance with HIPAA requirements.

EDUCATION

Bachelor of Science in Computer Science
Morgan State University

April 2010 - March 2014